TARUN SINGH GOHIL

AI Full Stack Developer | MERN · React.js · Node.js · Express.js · Python · PostgreSQL

Jaipur, Rajasthan, India | +91 8505015184 | tarungohil80@gmail.com

linkedin.com/in/tarun-singh-gohil | github.com/tarunsinghgohil

PROFESSIONAL SUMMARY

AI Full Stack Developer with over 6 years of experience building production web applications using the MERN stack (MongoDB, Express.js, React.js, Node.js), TypeScript, Python, Django, PostgreSQL, REST APIs, and modern UI/UX technologies. Practical Generative AI / AI engineering exposure across OpenAI GPT/LLM integrations, prompt engineering, RAG-style retrieval, embeddings/vector search, OCR/document understanding, and speech-to-text workflows. Currently working as a Lead Developer on a multi-tenant, AI-powered Hospital Information Management System, owning React.js architecture while integrating Node.js/Python backend services, authentication, RBAC, APIs, databases, and AI-enabled workflows. Strong in reusable architecture, scalable application design, reliable APIs, debugging, production problem solving, cloud/container fundamentals, and Agile/Scrum delivery.

PROFESSIONAL EXPERIENCE

Lead Developer | Growit.ai India Private Limited, Jaipur | Aug 2025 – Aug 2026

Full-stack (MERN) development and frontend ownership for AI-powered enterprise SaaS products.

- Led full-stack development of a scalable, multi-tenant, AI-powered Hospital Information Management System (HIMS/HMS), working hands-on across React.js/MERN frontend architecture, Node.js/Express.js services and Python/Django backend integrations, while contributing to OpenAI GPT/LLM-powered document understanding, summarization, content generation and assisted data-entry workflows.
- Architected enterprise applications across MERN and Python-based backend technologies, using MongoDB, Express.js, React 19, Node.js, Python, Django, Django REST Framework, TypeScript, JavaScript, Redux Toolkit, React Router and Tailwind CSS.
- Built and integrated Python/Django REST Framework APIs for enterprise healthcare workflows, including serializers, viewsets, middleware, authentication, permissions, ORM queries, migrations, validation, pagination, filtering and request/response handling.
- Developed and integrated Node.js/Express.js REST APIs with MongoDB/Mongoose, implementing backend services, request validation, authentication workflows, pagination, error handling, business logic and data operations for large-scale enterprise features.
- Worked across Python/Django/DRF and Node.js/Express.js services, designing API contracts, payload structures, response models and frontend integrations while troubleshooting authentication, data, API and deployment issues throughout the application lifecycle.
- Delivered 9+ complex healthcare modules end to end, including Patient Management, OPD, IPD, Emergency, Laboratory, Radiology, Pharmacy, Billing, Inventory, Analytics, Reporting and NABH-related workflows, covering business logic, APIs, data flows and production-ready UI.
- Contributed to AI/OCR-enabled document workflows, connecting document extraction and LLM-assisted capabilities with clinical, reporting and billing-oriented application flows, including OCR, document understanding and speech-to-text workflows using Whisper/Faster-Whisper where applicable.
- Implemented RAG-style retrieval workflows covering document ingestion, chunking, embeddings, similarity/vector search and grounded response generation, integrating retrieved context into Python/API-driven business workflows.
- Exposed AI capabilities through Python/Django REST services using structured JSON outputs, request validation, error handling, retries, timeouts, usage controls and safe fallback patterns for reliable downstream application integration.
- Designed reusable React 19/TypeScript component architecture including forms, dashboards, data tables, filters, validation flows and role-based interfaces using Redux Toolkit, Context API, Tailwind CSS, Material UI and Bootstrap.
- Implemented secure JWT authentication, protected routes, RBAC and tenant-aware authorization, supporting multiple hospital organizations, user roles and permission-driven business workflows.
- Worked with MySQL and MongoDB/Mongoose for application data operations, schema design, indexing, query optimization and caching strategies across data-intensive healthcare workflows.

- Improved application performance and scalability through code splitting, lazy loading, memoization, caching, efficient rendering, API optimization and bundle optimization, using Chrome DevTools, Lighthouse and performance profiling to identify bottlenecks.
- Managed Git-based development and deployment workflows including branching, pull requests, code reviews, release coordination, Docker-based development environments and deployment support across cloud environments.
- Collaborated with product managers, UI/UX designers, QA engineers and backend developers in Agile/Scrum teams, participating in technical discussions, debugging, production issue resolution and release activities.

UI Developer | Formidium India Private Limited, Jaipur | Feb 2022 – Jul 2025

- Developed and maintained production-grade MERN and Python-supported enterprise SaaS/fintech applications, working across React.js, TypeScript, Node.js, Express.js, Python, Django and REST APIs.
- Built and integrated Python/Django-based backend services and REST APIs for business workflows, including request validation, authentication, ORM-based database operations, data processing and third-party system integrations.
- Developed Python utilities and backend workflows for data extraction, transformation, validation and reporting, helping automate repetitive business processes and improve consistency of data-driven operations.
- Worked on AI/LLM-enabled application capabilities and RAG-oriented workflows, integrating retrieved context, structured model outputs and AI-assisted functionality into existing React, Node.js and API-driven business applications.
- Applied prompt engineering, document-understanding and AI-assisted automation patterns for reporting and business-data workflows, including response validation, error handling and practical model-integration debugging.
- Developed and maintained scalable React/TypeScript applications using reusable components, custom hooks, Redux, Redux Toolkit and Context API for dashboards, operational tools, business workflows and customer-facing products.
- Built and integrated Node.js/Express.js REST services with React applications and MongoDB, working across API contracts, data structures, asynchronous workflows, error handling and backend/frontend communication.
- Worked with backend teams on API design, pagination, response handling, database/query performance and production debugging across frontend, backend and data layers.
- Developed responsive and accessible interfaces from Figma designs using React.js, TypeScript, Tailwind CSS, Material UI, Bootstrap, HTML5 and CSS3, maintaining consistent UI/UX across desktop, tablet and mobile devices.
- Built analytics and operational dashboards using Chart.js and Recharts, presenting KPIs, financial information, operational metrics and business insights sourced from backend APIs.
- Managed and optimized 12+ production websites and applications through code splitting, lazy loading, caching, API optimization, efficient rendering and Webpack/build optimization.
- Integrated custom and third-party APIs using Axios, REST/JSON and asynchronous JavaScript, implementing dynamic data flows, validation, error handling and reliable frontend-backend communication.
- Worked across MySQL/MongoDB-backed applications and supported database-driven workflows, query optimization and application-level data handling.
- Collaborated with backend engineers, QA, UI/UX designers and product managers in Agile/Scrum teams, contributing to Git-based development, code reviews, testing, debugging, deployment and continuous improvement.
- Leveraged AI-assisted development tools including GitHub Copilot, Claude, Cursor and ChatGPT to improve coding productivity, debugging, documentation and feature delivery while maintaining clean, reusable and production-ready code.

Web Developer | i3Techs, Jaipur | Nov 2020 – Feb 2022

- Developed and maintained 50+ production websites and web applications, with a strong focus on HTML5, CSS3, JavaScript, Bootstrap and React.js, building responsive, performant and user-friendly interfaces across multiple client projects.

- Converted PSD and Figma designs into pixel-perfect, responsive production interfaces, applying mobile-first development, reusable UI patterns, semantic HTML, modern CSS practices and component-based frontend development.
- Built reusable React.js UI components, layouts and frontend structures for selected web applications, improving consistency, maintainability and development efficiency across projects.
- Developed responsive landing pages, product pages, business websites and customer-facing web interfaces, ensuring accessibility, usability, SEO-friendly markup and consistent experiences across desktop, tablet and mobile devices.
- Worked on API-driven frontend integrations, consuming backend data and collaborating on server-side workflows to connect web interfaces with application functionality and dynamic content.
- Customized WordPress themes and plugins and developed customer-facing websites using Elementor, while implementing responsive layouts, SEO best practices, reusable patterns and client-specific functionality.
- Ensured cross-browser and cross-device compatibility, identifying and resolving UI, rendering, responsiveness and browser-specific issues during development, testing and production maintenance.
- Designed and implemented modern visual interfaces using Adobe Photoshop and Adobe XD, collaborating with designers and clients to translate business requirements, branding guidelines and design specifications into production-ready experiences.
- Applied frontend engineering best practices across projects, including responsive design, accessibility, reusable code, CSS organization, debugging, Git-based version control and deployment support.

Web Development Trainee | i3Techs, Jaipur | May 2019 – Jul 2019

- Gained practical exposure to web development, UI/UX design workflows, responsive web design and modern frontend development practices across multiple client projects.
- Designed web interfaces, graphics and mockups using Adobe XD and Photoshop, focusing on visual consistency, typography, spacing, usability and alignment with project requirements.
- Contributed to UI/UX design and frontend implementation activities, developing an understanding of responsive layouts, design systems, usability principles and production-oriented web workflows.

KEY PROJECTS

1. Zonov — Multi-Tenant Hospital Management System (HIMS/HMS)

React.js, React 19, TypeScript, Redux Toolkit, Tailwind CSS, Node.js, Express.js, MongoDB, Mongoose, Django REST Framework, REST APIs, JWT, RBAC, OpenAI GPT, RAG, AI/OCR, Whisper/Faster-Whisper

2. RetailX ERP — POS Billing, Inventory & Reporting

React.js, TypeScript, Redux Toolkit, Tailwind CSS, Node.js, Express.js, MongoDB, REST APIs

3. Charge My EV — EV Charging Platform

React Native, React.js, Node.js, Express.js, MongoDB, REST APIs

TECHNICAL SKILLS

Python & Backend: Python 3.x, Python 3.11, OOP, abstraction, encapsulation, inheritance, polymorphism, decorators, decorator chaining, generators, iterators, comprehensions, slicing, mutability, context managers, custom exceptions, type hints, async/await, threading vs multiprocessing, clean-code practices; Django, Django REST Framework (DRF), Django ORM, serializers, viewsets, middleware, authentication, permissions, migrations, signals, validation, pagination, filtering, request/response handling; FastAPI, Pydantic, dependency injection, auto-generated API documentation, Flask, Celery-style background and asynchronous task processing.

Generative AI / LLM / ML / NLP: OpenAI GPT models, Azure OpenAI concepts, AWS Bedrock concepts, Google Vertex AI concepts, prompt engineering, prompt templates, system/role prompts, few-shot examples, structured JSON outputs, function/tool calling, RAG workflows, document ingestion, document chunking, embeddings, similarity/vector search, grounded generation, token and cost optimization, LLM evaluation, guardrails, scikit-learn, NumPy, Pandas, NLP concepts, computer vision, OCR, Hugging Face, document understanding, Whisper, Faster-Whisper and speech-to-text workflows.

Agentic AI: LangChain, LangGraph, CrewAI, AutoGen and Google ADK working knowledge; agent routing, tool calling and orchestration, multi-agent workflows, memory and state handling, workflow routing, human-in-the-loop review and approval patterns.

Frontend: React.js, React 19, JavaScript ES6+, TypeScript, React Hooks, component lifecycle, reusable component architecture, React Router, Next.js, Redux, Redux Toolkit, React Redux, Context API, Zustand, TanStack Query (React Query), form and API state management, Angular fundamentals, HTML5, CSS3, responsive web design, accessibility-aware implementation and cross-browser compatibility.
UI / Styling / Design: Tailwind CSS, Material UI (MUI), Bootstrap, Sass, LESS, jQuery, Radix UI, Lucide React, React Icons, Figma, Adobe XD, UI/UX principles, design systems, component library patterns, Figma-to-UI implementation, Figma/PSD-to-pixel-perfect UI development, wireframing and accessibility-focused UI development.
APIs & Architecture: REST APIs, RESTful API architecture, REST/JSON, API versioning, GraphQL, Axios, asynchronous programming, microservices concepts, JWT authentication, Role-Based Access Control (RBAC), multi-tenancy, pagination, validation, filtering, error handling, retries, timeouts, rate limiting, webhooks, WebSockets, third-party API integrations, payment integrations, healthcare technology integrations and WhatsApp Business API.
Databases & Data: MongoDB, Mongoose, MySQL, PostgreSQL, Redis; SQL joins, subqueries, normalization, indexing, transactions, schema and data modelling, ORM-based database operations, caching strategies and query optimization.
Cloud / DevOps / Infrastructure: AWS, AWS Lambda, EC2, S3, CloudWatch, IAM basics, serverless concepts and concurrency limits, Azure fundamentals, GCP fundamentals, Google Cloud, Docker, Docker Compose, Linux, Git, GitHub, GitLab, CI/CD workflows, environment and release management, deployment workflows, cloud deployment fundamentals and Vercel.
Build, Tooling & Testing: Vite, Webpack, Babel, npm, Yarn, ESLint, Chrome DevTools, browser debugging, debugging, performance profiling, Postman, pytest, unittest, fixtures and mocking, Jest, React Testing Library, Vitest, Cypress, Playwright and End-to-End (E2E) testing.
Performance / SEO: Code splitting, lazy loading, memoization, caching, bundle optimization, API optimization, Core Web Vitals, Lighthouse, performance profiling, responsive optimization and on-page Search Engine Optimization (SEO).
CMS & Platforms: WordPress, WordPress theme and plugin customization, Elementor, HubSpot CRM and HubSpot CMS.
AI Automation & Engineering Tools: OpenAI GPT integrations, OCR-enabled document workflows, speech-to-text workflows, Deepgram Speech-to-Text, Whisper, Faster-Whisper, AI-assisted development using GitHub Copilot, Claude, Cursor and ChatGPT.
Version Control & Engineering Practices: Git, GitHub, GitLab, Git-based version control, code reviews, Agile, Scrum, software debugging, production issue triage, release management, deployment support, frontend architecture, backend architecture, component architecture, design-system architecture, API integration patterns and authentication & authorization.

EDUCATION

Bachelor's Degree in CSE| Global Institute of Technology, Jaipur — August 2016 - August 2020
Senior Secondary Education (Science) | Anand Public School, Sawai Madhopur — August 2016

LANGUAGES

English, Hindi